

LILONGWE UNIVERSITY OF AGRICULTURE AND NATURAL RESOURCES

(LUANAR)

ASSIGNMENT SUBMISSION-ODL



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PROGRAMMEE	BABS
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QUESTION #1

To calculate by any method the population growth rate during the period(1959 to 69) for each town.

SOLUTIONS

$$\text{Population Growth Rate} = \frac{\text{Population in 1969} - \text{Population in 1959}}{\text{Population in 1959}}$$

$$\text{Jinja : Population Growth Rate} = \frac{47298-29741}{29741} \times 100 = \mathbf{58.83\%}$$

$$\text{Mbale : Population Growth Rate} = \frac{23539-13569}{13569} \times 10 = \mathbf{73.63\%}$$

$$\text{Entebbe : Population Growth Rate} = \frac{21176-10941}{10941} \times 100 = \mathbf{93.4\%}$$

$$\text{Gulu : Population Growth Rate} = \frac{19707-4770}{4770} \times 100 = \mathbf{3.12\%}$$

$$\text{Tororo : Population Growth Rate} = \frac{14136-6365}{6365} \times 100 = \mathbf{1.22\%}$$

$$\text{Masaka : Population Growth Rate} = \frac{13008-4782}{4782} \times 100 = \mathbf{1.72\%}$$

$$\text{Soroti : Population Growth Rate} = \frac{12330-6645}{6645} \times 100 = \mathbf{85.29\%}$$

$$\text{Kabala : Population Growth Rate} = \frac{11239-10919}{10919} \times 100 = \mathbf{2.88\%}$$

$$\text{Aura : Population Growth Rate} = \frac{11099-4645}{4645} \times 100 = \mathbf{1.38\%}$$

$$\text{Mbarara : Population Growth Rate} = \frac{10273-3844}{3844} \times 100 = \mathbf{1.67\%}$$

QUESTION #2

SOLUTIONS

- a) Calculating average annual rate of increase, by using this formula

$$\text{Average Annual Rate of Increase} = \left\{ \frac{\text{population in 1969}}{\text{population in 1959}} \right\}^{\left[\left(\frac{1}{10} \right) - 1 \right]} \times 100$$

$$\text{Average Annual Rate of Increase} = \left\{ \frac{9,456,466}{6,449,558} \right\}^{\left[\left(\frac{1}{10} \right) - 1 \right]} \times 100 = \mathbf{3.16\%}$$

- b) Estimating the population in 1964 and 1967 by using average annual rate of increase to calculate:

$$\text{Population in 1964} = 6,449,558 \times (1 + 0.0316)^5 = \mathbf{7,468,125}$$

$$\text{Population in 1967} = 6,449,558 \times (1 + 0.0316)^8 = \mathbf{8,309,413}$$

- c) Estimating the population in 1975 by using average annual rate of increase formula;

$$\text{Population in 1965} = 6,449,558 \times (1 + 0.0316)^{15} = \mathbf{12,815,611}$$

- d) To calculate number of years that will double itself, by using the rule of 70.

$$\text{Number of years to double} = \frac{70}{3.16} = \mathbf{22.15 \text{ years}}$$

QUESTION #3

Calculating the rate of growth by regions in Ghana during the interval periods by using this formula:

SOLUTIONS

$$\text{Rate of Growth} = \frac{\text{population in 1970} - \text{population in 1960}}{\text{population in 1960}}$$

Western

$$\text{Rate of Growth} = \frac{203,137 - 154,612}{154,612}$$

$$\underline{\text{Rate of Growth} = 0.31}$$

Central

$$\text{Rate of Growth} = \frac{247,373 - 210,411}{210,411}$$

$$\underline{\text{Rate of Growth} = 0.17}$$

Greater Accra

$$\text{Rate of Growth} = \frac{691,183 - 393,383}{393,383}$$

$$\underline{\text{Rate of Growth} = 0.75}$$

Eastern

$$\text{Rate of Growth} = \frac{316,878 - 220,765}{220,765}$$

$$\underline{\text{Rate of Growth} = 0.43}$$

Volta

$$\text{Rate of Growth} = \frac{154,016 - 102,101}{102,101}$$

$$\underline{\text{Rate of Growth} = 0.51}$$

Ashanti

$$\text{Rate of Growth} = \frac{464,016 - 204,772}{204,772}$$

$$\underline{\text{Rate of Growth} = 1.26}$$

Brong Ahafo

$$\text{Rate of Growth} = \frac{214,392 - 91,491}{91,491}$$

$$\underline{\text{Rate of Growth} = 1.34}$$

Northern

$$\text{Rate of Growth} = \frac{142,845 - 69,063}{69,063}$$

$$\underline{\text{Rate of Growth} = 1.06}$$

Upper

$$\text{Rate of Growth} = \frac{62,027 - 32,576}{32,576}$$

$$\underline{\text{Rate of Growth} = 0.90}$$

QUESTION #4

Calculating the rate of population growth by districts in Uganda

$$\text{Population Growth Rate} = \frac{\text{Population in 1969} - \text{Population in 1959}}{\text{Population in 1959}} \times 100$$

West Mengo: $\text{Population Growth Rate} = \frac{803,605 - 508,410}{508,410} \times 100\% = 58.20\%$

East Mengo: $\text{Population Growth Rate} = \frac{846,986 - 606,694}{606,694} \times 100 = 39.57\%$

Masaka : $\text{Population Growth Rate} = \frac{637,018 - 440,180}{440,180} \times 100 = 44.70\%$

Mubende : $\text{Population Growth Rate} = \frac{329,998 - 215,739}{215,739} \times 100 = 53.00\%$

Teso : $\text{Population Growth Rate} = \frac{565,936 - 453,474}{453,474} \times 100 = 24.82\%$

Bugisu : $\text{Population Growth Rate} = \frac{415,827 - 311,474}{311,474} \times 100 = 33.47\%$

Bukedi : $\text{Population Growth Rate} = \frac{524,723 - 397,650}{397,650} \times 100 = 31.94\%$

Busonga : $\text{Population Growth Rate} = \frac{933,270 - 660,507}{660,507} \times 100 = 41.34\%$

Sebei : $\text{Population Growth Rate} = \frac{64,432 - 49,320}{49,320} \times 100 = 30.71\%$

$$\text{Karamoja : Population Growth Rate} = \frac{283,776-171,945}{171,945} \times 100 = \mathbf{65.20\%}$$

$$\text{Kigezi : Population Growth Rate} = \frac{646,726-493,444}{493,444} \times 100 = \mathbf{31.01\%}$$

$$\text{Ankole : Population Growth Rate} = \frac{859,113-529,712}{529,712} \times 100 = \mathbf{62.13\%}$$

$$\text{Toro : Population Growth Rate} = \frac{567,907-347,479}{347,479} \times 100 = \mathbf{63.40\%}$$

$$\text{Bunyoro : Population Growth Rate} = \frac{350,286-189,980}{189,980} \times 100 = \mathbf{84.42\%}$$

$$\text{West Nile: Population Growth Rate} = \frac{572,286-383,926}{383,926} \times 100 = \mathbf{49.11\%}$$

$$\text{Madi :-Population Growth Rate} = \frac{89,842-50,627}{50,627} \times 100 = \mathbf{77.47\%}$$

$$\text{Acholi: Population Growth Rate} = \frac{462,095-285,530}{285,530} \times 100 = \mathbf{61.86\%} \quad \text{Lango}$$

$$\text{:-Population Growth Rate} = \frac{502,637-352,943}{352,943} \times 100 = \mathbf{42.37\%}$$

$$\text{Uganda :-Population Growth Rate} = \frac{9,456,466-352,943}{352,943} \times 100 = \mathbf{46.60\%}$$